Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE – B Section
Course Code & Title: EE8301 Electrical Machines - I
Name of the Faculty member (s): Mr.E. Thangam

Innovative Practice Description

- **Unit / Topic:** Unit I / Hysteresis and Eddy Current losses
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO - 2
- **Activity Chosen:** Demonstration
- **Justification:**

The Hysteresis and Eddy Current losses depend on the types of cores. Since each type has its own property. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the solid and laminated cores which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 29,

Photographer: one student – Ms. T. Udaya (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about various types of cores in electrical machines for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the construction of cores for another 1 minute.
- ✓ Finally, I shown the stampings and laminated core for each student and explained.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 5	PO 10	PO12	PSO 1
CO1	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	Due to demonstration practice students understood about the core used in the electrical machines and the impact on electrical equipments performance and society. So it is correlated slightly.

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  Students told that it is good see all the parts of the machines individually and demonstration helps the students to understand the concepts easily.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Most of the students attended the question asked Internal Assessment Test – I retest.
3. The success of the activity was evaluated by asking the same question during transformer core construction (Unit – II) **– Around 80% of students answered.**
4. Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

- https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations
- Nagrath, I.J. and Kothari.D.P., Electric Machines', McGraw-Hill Education, 2004

Signature of Faculty Member **HOD**

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Innovative Practice Description

- **Unit / Topic: Unit II / Constructional Features of Transformer**
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO - 4
- **Activity Chosen:** Field Visit
- **Justification:**

The topic construction of transformer has different parts, since each part has used for separate purposes. So, I thought of conducting this field visit for making the students to give the constructional features of transformers which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

After teaching the concept, students were took to the RIT – Power House Transformer yard and ask them to think about the constructional features of transformers.

Total Strength is 31,

Photographer: one student - Mr. Nikil Dev (interested in photography)

Reporter: Myself

At the end of the visit (Last 5 minutes)

- ✓ I asked the students to think about various parts of transformers for 2 minutes.
- ✓ Then I told them to Pair with their neighbors and discuss about the construction of transformer for another 1 minute.
- ✓ Finally, I randomly asked students about the construction parts of transformers. (2 minutes)

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO 7	PO 8	PO 10	PO12	PSO 1	PSO 3
CO2	2	1	1	2	1	1	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 7**
	1
Justification for correlation	Due to field visit students can able to understand the impact of engineering solution on society (i.e.) transformer operation and failure impact on society. So it is correlated slightly.

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  Students enjoyed the field trip and especially boys asked more questions and clarified their doubts. Students asked, similar trips can be arranged in future for every unit of syllabus.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Students can able to attend the question even in the questions are in indirect form.
3. Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. In the visit mostly girls hesitate to asking the questions.
2. Time utilization for conducting activity.

References:

- Nagrath, I.J. and Kothari.D.P., Electric Machines', McGraw-Hill Education, 2004

- Daniel PAEZ and Luis Alberto RUBIO, Colombia, "The Use of Field Trips in the Context of Engineering Collaborative Teaching: Experiences of Hands-On Geomatics Activities in Colombia", Peer reviewed paper, 2015.

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Innovative Practice Description

- **Unit / Topic:** Unit III / Concepts in Rotating Machine
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO - 9
- **Activity Chosen:** Demonstration
- **Justification:**

The topic basic parts in electrical machines will vary machine to machine. Since each machine (A.C. and D.C.) has different parts. After teaching the concept, I thought of conducting this activity for making the students to give the knowledge about the different parts of different machines which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**

After teaching the various parts of A.C. and D.C. machines, I explained the various parts of A.C. and D.C. machine using stampings.

Total Strength is 31,

Photographer: one student – Mr. T. Udaya (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I explained the various parts of A.C. and D.C. machine using stampings for 5 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the various parts of electrical machines for another 3 minute.
- ✓ Finally, I asked the students to summarize each parts of the different machines. (2 minutes)

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 10	PO12	PSO 1
CO3	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	Due to demonstration practice, students understanding level about the constructional features of the electrical machines will increase. Also understand the slight change in the machine design how affect the society. So it is correlated slightly.

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  Students told that it is new for us to understand the various parts of electrical machines through this demonstration and the felt good about this demonstration.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society.
2. The assessment of effectiveness of the activity was felt when told most of the points.
3. While conducting the activity, I understood that the students can able to explain the various parts of electrical machines.
4. Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

- https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations
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Innovative Practice Description

- **Unit / Topic:** Unit IV / Constructional features of D.C. Machines
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO - 10
- **Activity Chosen:** Demonstration
- **Justification:**

The topic constructional feature of D.C. Machines is same for generator and motor. But the way of excitation is different. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the D.C. motor and D.C. generator which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 29,

Photographer: one student - Mr. T. Udaya (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I took the students to the electrical machines laboratory and asked the students to think about constructional features of D.C. Machines for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the constructional features of D.C. Machines for another 2 minute.
- ✓ Then, I shown the demo D.C. Machine for each student and explained. (8 minutes)
- ✓ Finally I asked each student to tell about constructional feature. (3 Minutes)

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO8	PO 10	PO12	PSO 1
CO4	3	2	1	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 7**
	1
Justification for correlation	Due to demonstration practice, students understanding level about the constructional features and operating principle of DC machine will increased. So it is correlated slightly.

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  Students felt good about this demonstration.

- ***Benefit of the practice:***

1. The assessment of effectiveness of the activity was felt when told most of the points.
2. While conducting the activity, I understood that the students can able to explain the core construction of electrical machines.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

- https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations
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